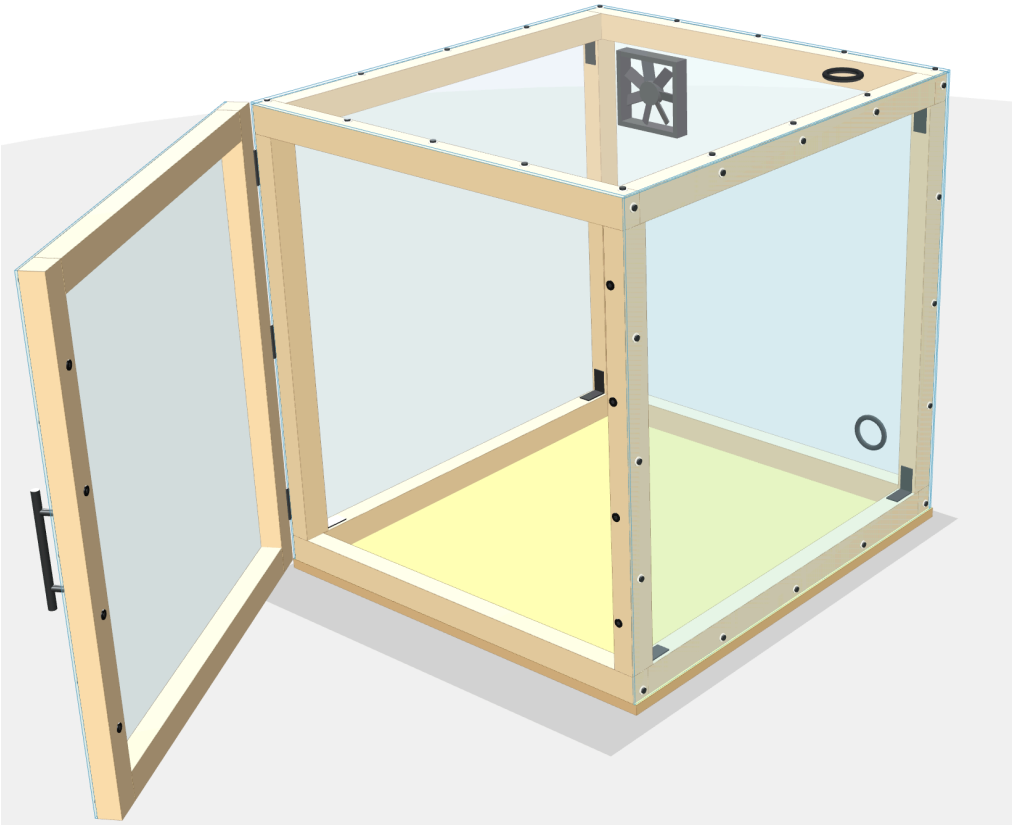
# Bambu Lab A1 Wooden Enclosure

Timber drawings • external feed (AMS lite outside) • plain-strip version



3D model view (door half open, printer hidden)

## General notes

1. All dimensions in millimetres (mm).
2. Timber: 30×40 strip, kiln-dried spruce or pine (8–12 % moisture), planed four sides, straight grain with few knots. Paulownia also works but holds screws poorly.
3. The timber shop only cuts to length: no grooves, no rebates, no holes. Square (90°) end cuts.
4. Tolerances: length ±1, section ±0.5.
5. Panels are 3 or 4 mm clear twin-wall PC lying on the outside of the frame on 3 mm foam tape, held by ST4×25 screws with M6 large flat washers, not glue (detail on sheet 4). The panel sizes are the same as for a 3 mm sheet, so the timber is unaffected.
6. Pre-drill Ø3 pilot holes for every wood screw, especially near strip ends; glue every wood-to-wood joint.
7. Pencil the part letter (A–E) on every piece.

## Sheet list


1. Overview
2. Cut list
3. Frame assembly views
4. Door frame and fixing details
5. Twin-wall panels and holes


| | |
|---|---|
| Frame (outside of wood) | **640 × 700 × 680** |
| Clear inside W×D×H | **580 × 640 × 640** |
| Finished size (door, feet) | **about 650 × 741 × 708** |
| 30×40 strip total | **10.0 m, 16 pieces (6 × 2 m)** |
| Clear panels | **3 or 4 mm clear twin-wall PC, 5 panels, 16 × Ø6 holes each (drilled by you)** |

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**Overview**

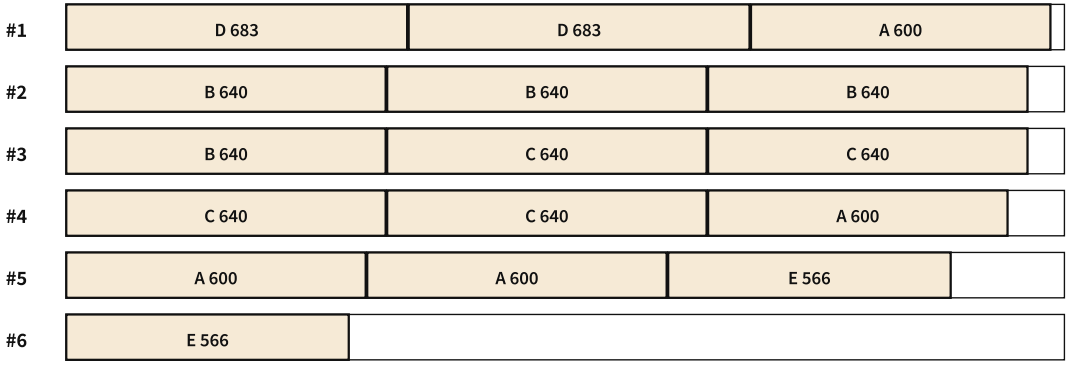
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 1 / 5 | — | mm | 2026-11-08 | v2 |

## Cut list • 30×40 strip (cut to length only)

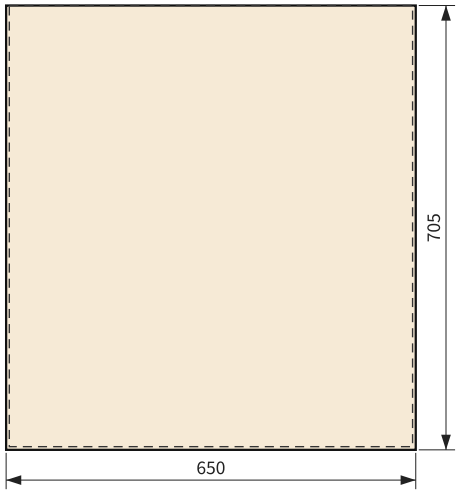
| ID | Part | Size | Qty | Location |
|---|---|---|---|---|
| **A** | Post | 30×40×600 | 4 | One per corner, 30 mm face to front/back, standing on the bottom frame |
| **B** | Top/bottom frame, front & back rail | 30×40×640 | 4 | Full length, 40 mm on edge |
| **C** | Top/bottom frame, side rail | 30×40×640 | 4 | Fits between the front and back rails |
| **D** | Door stile | 30×40×683 | 2 | Door left & right, full height |
| **E** | Door rail | 30×40×566 | 2 | Fits between the stiles |
| **F** | Base board | 650×705×15 | 1 | 15 mm solid-wood panel or plywood |

**Strip total 10 018 mm, 16 pieces; 1 base board.**

## Cutting plan • 2 m stock, 3 mm kerf




**F Base board 1:12**



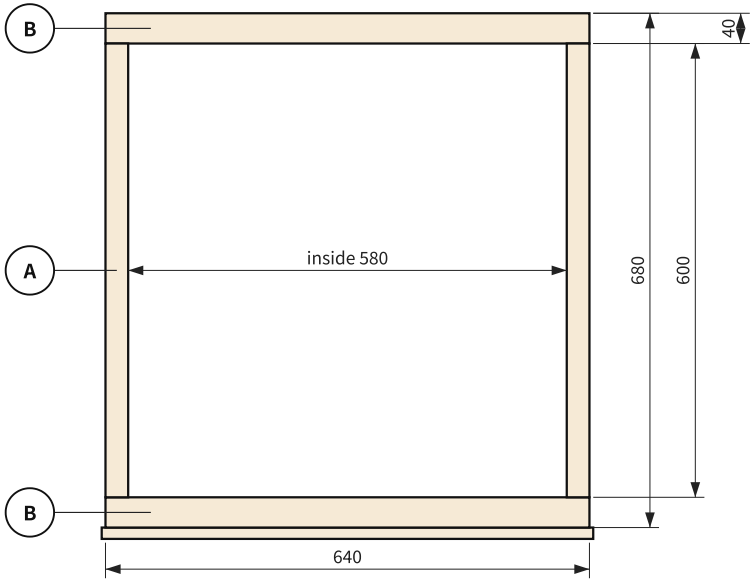

Dashed = frame footprint: flush at the front, 5 in at the left, right and back.

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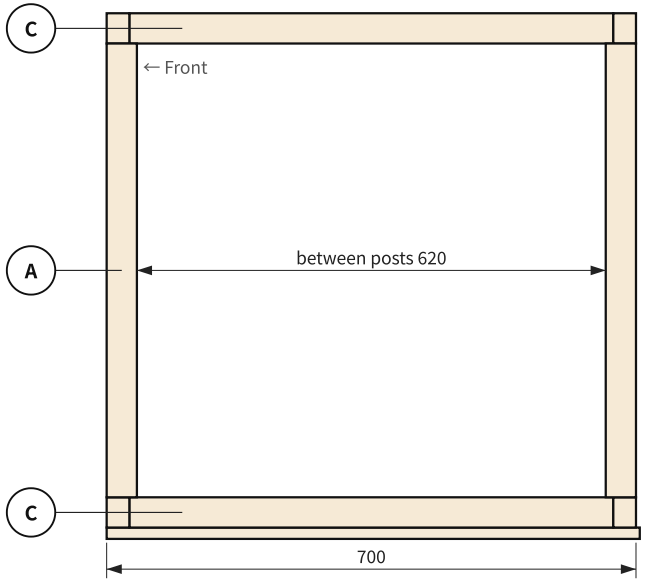
**Cut list**

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 2 / 5 | — | mm | 2026-11-08 | v2 |

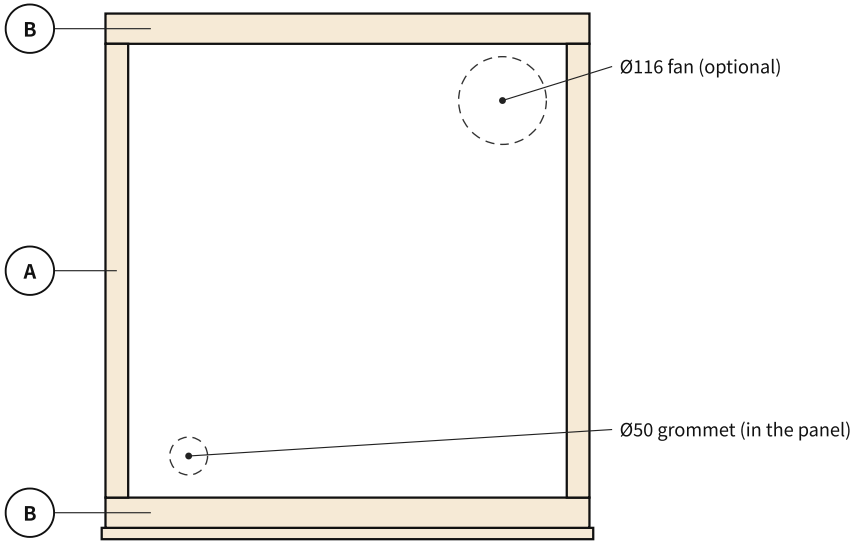
## Front view (door removed)



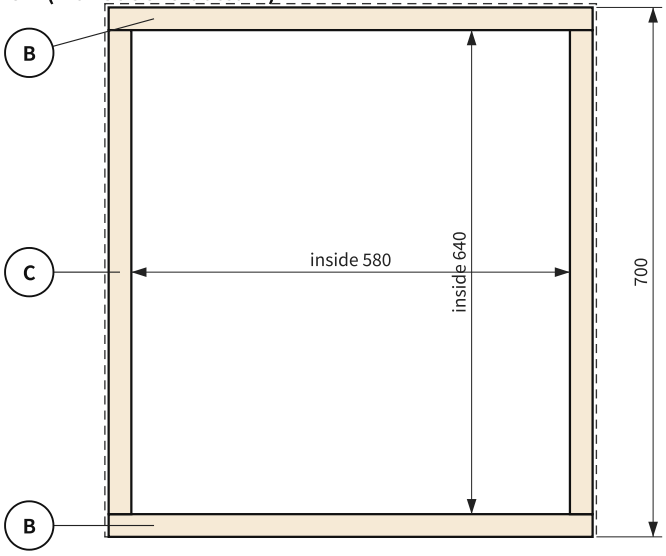

## Right side view (front on the left; left side mirrors it)




## Back view (from outside)




## Top view (front at the bottom)



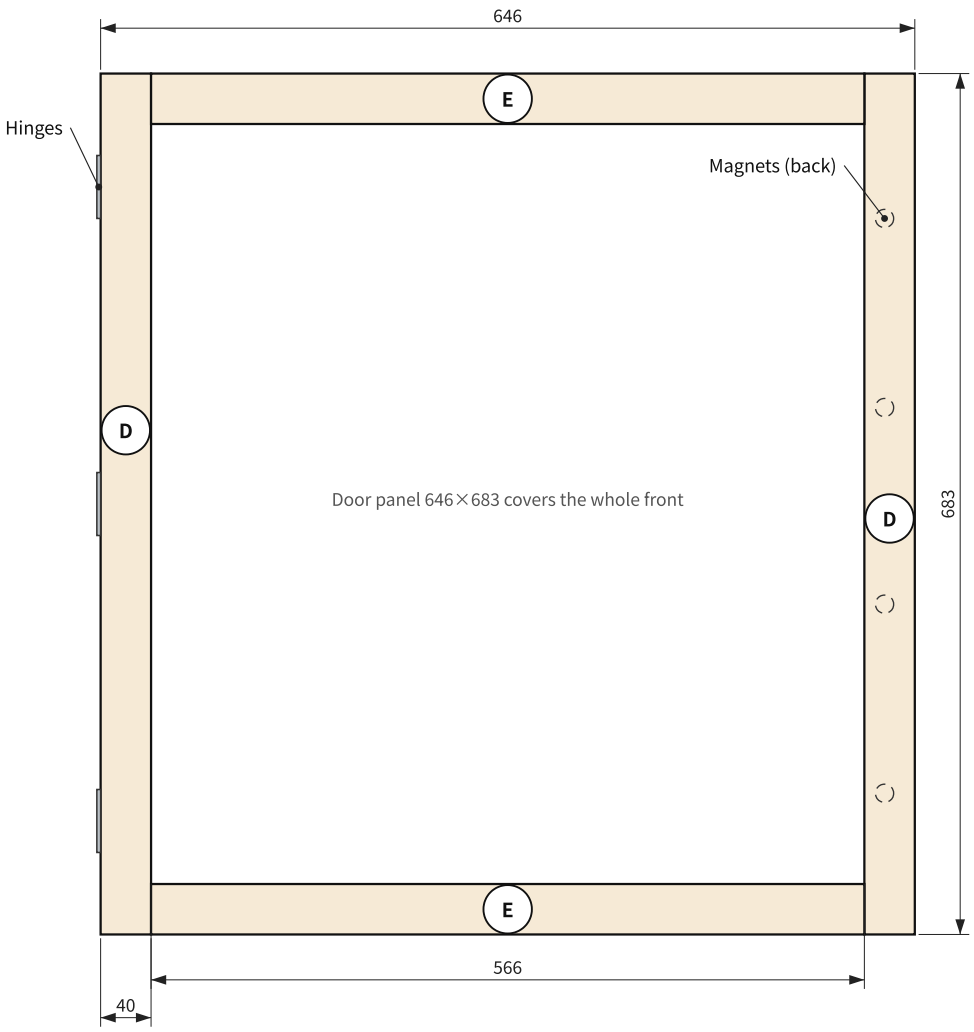

Rails B run full width, side rails C fit between them; posts A stand on the bottom frame and the top frame sits on the posts.

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# Frame assembly views

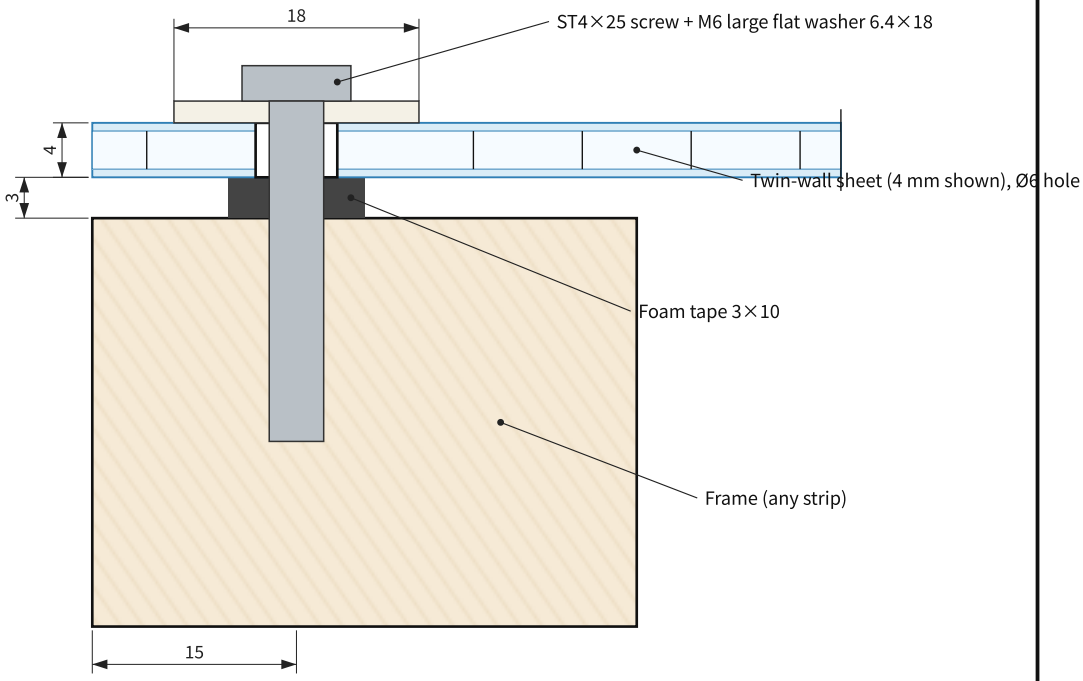
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 3 / 5 | 1:10 | mm | 2026-11-08 | v2 |

## Door frame, front view (from outside, scale 1:6)



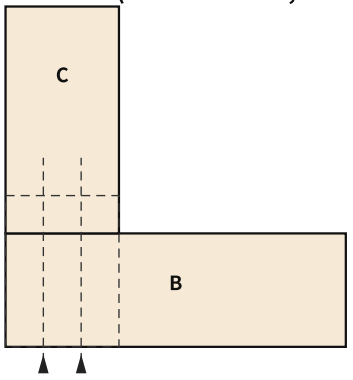

Stiles D run full height with rails E between them; glue each corner and drive two 4×60 screws through the stile into the rail end. The panel keeps the door square once fitted. Magnet pockets and hinges are DIY.

## Detail 1: twin-wall panel fixing (scale 1.8:1)




The Ø6 hole is 2 mm larger than the screw so the panel can expand. Use the large washer and stop as soon as it seats, then back off a quarter turn so the hollow sheet is not crushed; work from the middle of each edge outward. The screw bites about 16 mm into the wood.

## Detail 2: frame corner (bottom frame, front-left, plan 1:2)




Rail B runs through; side rail C butts against its inside face, fixed with two 4×50 screws through B into the end of C. Dashed = post A: two 4×60 screws from below into its end, plus an L bracket inside. The top frame is the same.

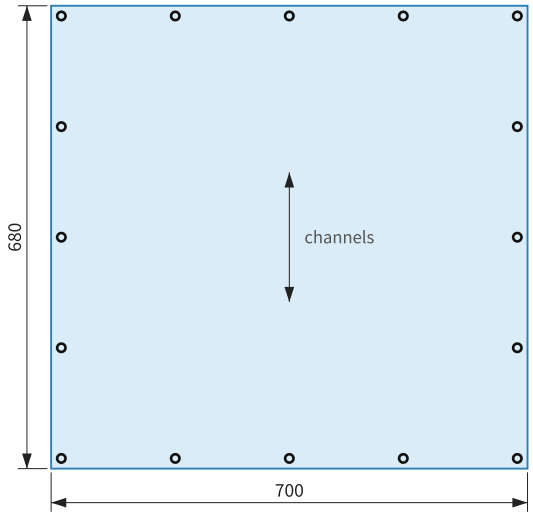
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# Door frame and fixing details

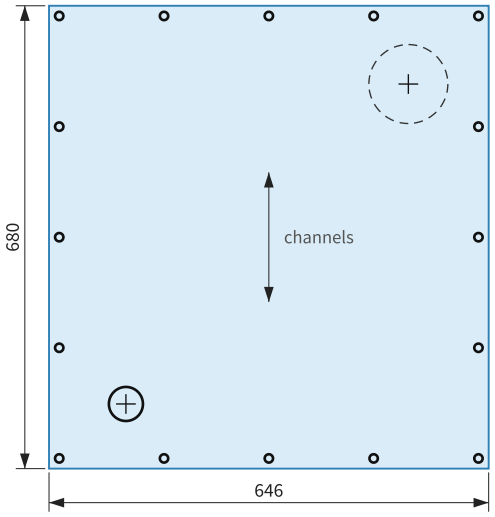
| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 4 / 5 | 1:6 / 2:1 / 1:2 | mm | 2026-11-08 | v2 |

## Twin-wall panels and holes (about 1:11, seen from outside)

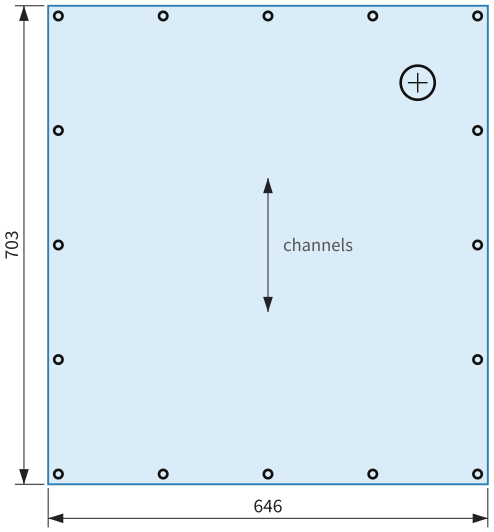
P1 Side panel (L/R) ×2



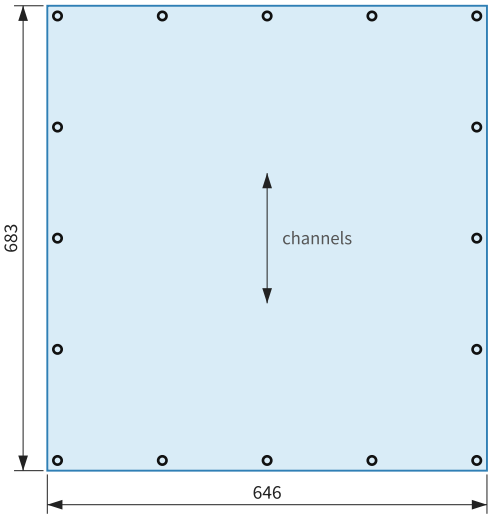

P2 Back panel ×1




P3 Top panel ×1




P4 Door panel ×1




| Panel | Size | Qty | Pitch (across / up) |
|---|---|---|---|
| **P1** | 700×680 | 2 | 167.5 / 162.5 |
| **P2** | 646×680 | 1 | 154 / 162.5 |
| **P3** | 646×703 | 1 | 154 / 168.25 |
| **P4** | 646×683 | 1 | 154 / 163.25 |

| Panel | Hole | Position (to hole centre, seen from outside) |
|---|---|---|
| **P2** | Ø50 grommet hole | 113 from the left edge, 95 from the bottom |
| **P2** | Ø116 fan hole (optional) | 118 from the right edge, 115 from the top |
| **P3** | Ø50 grommet hole | 103 from the right, 113 from the top (rear) edge |

- Each panel: 5 holes per edge (corners shared, 16 in all), Ø6, centres 15 from the edge. The supplier only cuts the sheets; you drill: sheet on scrap wood, clamped, low speed, film on; at assembly drill Ø3 pilot holes about 18 mm deep into the wood through the Ø6 holes.
- Material: 3 or 4 mm clear twin-wall PC, clear colour. Channel direction as the arrows show: vertical on all four, front to back on the top. Size tolerance ±1, film left on, edges need no polishing.
- Edge sealing: tape the two open edges of every sheet (top and bottom of the side, back and door panels, front and back of the top panel) with aluminium foil tape before fitting, about 6.7 m in all.
- Side panels cover the post sides, the back panel covers the side panels' rear edges, the top covers both. Grommet positions can be adjusted to the A1 power socket; drill the handle holes once you have the handle.

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# Twin-wall panels and holes

| Sheet | Scale | Units | Date | Rev. |
|---|---|---|---|---|
| 5 / 5 | ≈1:11 | mm | 2026-11-08 | v2 |